

Vadim Uvarov

ML Engineer | Data Scientist

PhD in Machine Learning

Male, 32

Industrial experience: 8 years

Location: Antwerp, Belgium



I am passionate about developing complex software using mathematical and/or machine learning models that helps business to solve its most significant problems.

Having experience both in academy and industry, I can formalize a loosely stated business task and conduct a research & development of a solution either independently or as a part of a team.

I also have an experience of leading a small data science team specialized in telecom geodata analysis.

Skills

Machine Learning

- **Classification** (venue mapping)
- **Regression** (retail site selection)
- **Time-series forecasting** (energy demand forecasting, solar energy forecasting, on-shelf availability detection)
- **Hidden Markov models** (map matching)
- **Clustering** (geo locations)
- **MLflow** (organizing training pipelines)

Deep Learning

- **TensorFlow, TensorFlow Lite** (on-device ML)
- **PyTorch** (energy demand forecasting)

Data visualization

- **matplotlib, seaborn, plotly**

Statistics

- **Hypothesis testing** for feature selection

Python

- **Pandas, Scikit-Learn** for data science
- **numpy / scipy** for algo implementation
- **pytest** for unit testing
- **commitizen, ruff, mypy** for clean code

Contacts

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Languages

English / C1 (advanced)

Dutch / A2 (elementary)

Russian / C2 (native)

Cloud Deployment

- **AWS, Kubernetes (K8s), Pulumi, Docker**
- **Gitlab CI/CD**

Software Engineering

- **Git**, unit and integration **testing**, code review, style guide, production-ready code, **Jira, Asana**
- Reactive programming: **ReactiveX**
- Observability: **Grafana, Prometheus**

Big Data

- **Postgres, Teradata, Hadoop, Spark**
- **SQL**: joins, aggregates, window functions

Algorithms and Data Structures

- Computational complexity analysis
- Basic graph algorithms (BFS, Dijkstra)
- Recursion, tree traversing

Research

- Scientific method
- Article writing and presenting
- Defended a **PhD** thesis

Work Experience



ML engineer – Centrica [energy company] – Antwerp, Belgium

Apr 2023 – Oct 2025

Centrica Business Solutions helps organisations to develop smart energy solutions.

I worked as a quantitative developer (a blend of a software engineer, data scientist and ML engineer).

- Designed, developed and deployed an **electric vehicle (EV) charging controller** from scratch. This enabled thousands of residential customers in the UK to save on their electricity bills and Centrica to reduce energy costs, making it the cheapest way of EV charging in the UK.
- Designed, developed and deployed a residential **energy demand forecaster**. It allowed to increase the savings more than 15% in comparison to the baseline.
- Designed, developed and deployed a physical-based model for **solar energy forecasting**. This significantly increased accuracy (vs the baseline) but also kept the results interpretable.



Data Scientist – Sentiance [AI startup] – Antwerp, Belgium

May 2021 – Jan 2023

Sentiance develops a Mobile SDK, which can be embedded by its clients into their mobile apps and provide them with insights into their users' motion data. I worked on the lifestyle component. It was decided to migrate the solution from backend to the on-device (edge computing).

- developed a **venue type mapping** model with **Python** and **TensorFlow** for on-device usage, that allows to predict, which type of venue a user visited (shop/restaurant/bar, etc.)
- applied **transfer learning** technique to increase the accuracy of the **venue type mapping** model
- ensured that the model training pipeline was tracked in **MLflow** to achieve reproducible results
- used **TensorFlowLite** to develop a multiplatform **on-device** implementation of the algorithm

This allowed for a blazingly fast on-device inference which also increased users' privacy.



Data Scientist -> Senior Data Scientist – Tele2 Russia [telecom] – Moscow, Russia

Jun 2018 – Apr 2021

At Tele2 telecom company I worked (first as individual contributor and then as a lead) of geodata team.

- enhanced subscribers' **track processing** for subscriber geolocation using triangulation technique
- significantly improved subscribers' **point-of-interest detection** (like home or work) accuracy
- implemented **map matching** solution in **Scala Spark** (for scalability) to snap millions of subscribers' geolocation to a particular street/road segment (here I applied findings from my PhD thesis).

This enhanced accuracy, reduced time-to market and provided ML features for the other teams.



Data Scientist – Magnit [food retailer] – Krasnodar, Russia

May 2017 – May 2018

I worked on a **retail site selection** problem: predict the shop potential revenue based on its coordinates.

Another project was **on-shelf availability** prediction system (based on time-series forecasting, not computer vision) to detect and promptly signalize that a particular product should be restocked.

Education



Novosibirsk State Technical University – PhD in Machine Learning

2015-2019

My research subject was “Incomplete sequence analysis using hidden Markov models”. I have defended my PhD thesis in February 2020.



Novosibirsk State Technical University – MSc in Applied Mathematics and Computer Science

2010-2015

The subject of my graduate thesis was “HMM implementation using GPU parallel computing”.

Recent Courses taken and Books read

Udemy course: [AWS Certified Developer Associate 2025 DVA-C02](#).

Book: “Designing Data-Intensive Applications” by Martin Klepmann.

Book: “Designing Machine Learning Systems” by Chip Huyen.

Coursera specialization (watched lectures): [Deep Learning](#) by DeepLearning.AI.

Coursera specialization: [machine learning and data science](#) by Yandex.

Course (watched lectures): [CS231n](#) Convolutional Neural Networks for Visual Recognition by Andrej Karpathy.

I’ve taken Dutch courses, which allowed me to achieve a certified A2 level in Dutch. I’m reading books and practice conversations in Dutch and to further advance my language skills.

My projects, articles, speeches

Research papers

“On-Device Deep Learning Location Category Inference Model” by Gadzhi Musaev, Kevin Mets, Rokas Tamosiunas, **Vadim Uvarov**, Tom De Schepper, Peter Hellinckx (Belgium, 2022):

https://bnaic2022.uantwerpen.be/wp-content/uploads/BNAICBeNeLearn_2022_submission_6606.pdf

My Research Gate profile: https://www.researchgate.net/profile/Vadim_Uvarov

Medium article “Feature selection using hypothesis testing”

https://medium.com/@vadim_uvarov/feature-selection-using-statistical-testing-2d8e7b5e27b8

Work Project “Map Matching of Cellular Data using hidden Markov models”

My Data Fest conference speech about the technical aspects of this project:

<https://youtu.be/kdyThl4CZT0> (in Russian with English subtitles, studio record).

Scientific paper about this project, written in collaboration with my colleagues (in Russian):

<http://media-publisher.ru/wp-content/uploads/2019/07/Nom-7-2019-s.pdf> (page 32).

PhD thesis “Incomplete sequence analysis using hidden Markov models”

Thesis (in Russian): www.nstu.ru/files/dissertations/dissertaciya_1574424031_157443103786.pdf

Video of my thesis defense (in Russian): <https://youtu.be/Qk9JDMm0nM>

Code and experiments: https://github.com/long-johnson/HMM_Python

Pet project: street signs character recognition (computer vision, deep learning)

To complement [CS231n](#) course videos with practice, I decided to implement a convolutional neural network for classifying images of Latin letters (capital and lowercase) and digits in the wild (Chars74k dataset). I used Python, Theano backend, Lasagne library for CNN construction, Keras for data augmentation and OpenCV for image preprocessing ([code on GitHub](#)). I achieved 83% accuracy score and 14-th place in [Kaggle training competition](#).

Hobbies

To let my mind unwind after work, I play padel, go to gym and explore Belgium. I play electric guitar and enjoy video games. I like travelling and getting familiar with foreign cultures.